

A 2-partial isometry whose canonical generalized inverse cannot be Moore–Penrose for any Hilbert inner product

Autonomous mathematical discovery run `fa_banach_001`

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Status. Candidate full negative answer, likely valid pending expert review.

1 Source question

Aouichaoui, Buchała, and Garcia observe that if T is a 2-partial isometry, then

$$S = 2T^* - T^{*2}T$$

is a generalized inverse of T , with ST an orthogonal projection for the original inner product. Their Question 6 asks whether this fixed operator S is the Moore–Penrose inverse of T with respect to *some* Hilbert inner product on the same space [1, Question 6, PDF p. 17].

Garcia and Sherman proved that $A \in M_n$ is similar to a partial isometry if and only if (a) $\sigma(A) \subseteq \mathbb{D}^-$; (b) if $\zeta \in \sigma(A) \cap \mathbb{T}$, then its algebraic and geometric multiplicities are equal; and (c) $\dim \mathcal{N}(A) \geq \dim \mathcal{N}(A - \lambda I)$ for each $\lambda \in \sigma(A) \cap \mathbb{D}$ [21]. Does an analogous characterization exist for m -partial isometries?

Question 5. Can m -partial isometries on finite-dimensional Hilbert spaces be concretely classified up to similarity?

The condition $T\beta_2(T) = 0$ can be written as

$$T(2T^* - T^{*2}T)T = T,$$

so $S = 2T^* - T^{*2}T$ is a generalized inverse for T and ST is the orthogonal projection onto $\mathcal{R}(T^*)$. Moreover, $STS = S$, so S is almost the Moore–Penrose inverse of T . However, TS need not be selfadjoint: if

$$T = \begin{bmatrix} a & 0 \\ 1 & 0 \end{bmatrix}$$

with $|a|^2 = \frac{1+\sqrt{5}}{2}$, then T is a 2-partial isometry and

$$TS = \begin{bmatrix} -1 & 2a \\ \bar{a} \frac{1-\sqrt{5}}{2} & 2 \end{bmatrix}$$

is not selfadjoint. All is not lost, however. The term “selfadjoint” above refers to the standard inner product on $\mathbb{C}^2$. Since different inner products give rise to different adjoints, the next question is natural.

Question 6. If T is a 2-partial isometry, is $2T^* - T^{*2}T$ the Moore–Penrose inverse of T with respect to some inner product?

We give a negative answer. In fact, the 2×2 matrix displayed immediately before the question in the source already supplies a counterexample.

2 Proof intuition

Changing the inner product on $\mathbb{C}^2$ amounts to choosing a positive definite Hermitian matrix G . A fixed operator A is selfadjoint for the new inner product exactly when $A^*G = GA$.

For the source's example, the two idempotents in the Penrose equations are

$$ST = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad TS = \begin{pmatrix} -1 & 2a \\ a\delta & 2 \end{pmatrix}, \quad \delta < 0.$$

Making ST selfadjoint forces G to be diagonal. Making TS selfadjoint with respect to the same diagonal $G = \text{diag}(x, y)$ would then force $2x = \delta y$. The left side is positive and the right side negative. Thus no Hilbert inner product can make both Penrose idempotents selfadjoint.

3 Counterexample

We first record the standard change-of-inner-product test.

Lemma 1. *Let $T, S \in M_n(\mathbb{C})$ satisfy $TST = T$ and $STS = S$. The operator S is the Moore–Penrose inverse of T for the inner product*

$$\langle x, y \rangle_G = \langle Gx, y \rangle, \quad G = G^* > 0,$$

if and only if

$$(TS)^*G = G(TS) \quad \text{and} \quad (ST)^*G = G(ST).$$

Proof. The adjoint for this inner product is $A^\sharp = G^{-1}A^*G$. The first two Penrose equations are already the two assumed generalized-inverse identities. The remaining two say precisely that TS and ST are $\sharp$ -selfadjoint, which is equivalent to the displayed matrix equations. $\square$

Theorem 2. *Question 6 of [1] has a negative answer. More explicitly, let*

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad a = \sqrt{\varphi}, \quad \delta = 1 - \varphi = \frac{1 - \sqrt{5}}{2},$$

and define

$$T = \begin{pmatrix} a & 0 \\ 1 & 0 \end{pmatrix}.$$

Then T is a 2-partial isometry, but

$$S = 2T^* - T^{*2}T$$

is not the Moore–Penrose inverse of T with respect to any Hilbert inner product on $\mathbb{C}^2$.

Proof. Since $a^2 = \varphi$ and $\varphi^2 = \varphi + 1$, direct multiplication gives

$$T^*T = \begin{pmatrix} \varphi + 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad T^{*2}T^2 = \begin{pmatrix} \varphi^2 + \varphi & 0 \\ 0 & 0 \end{pmatrix}.$$

Consequently,

$$\beta_2(T) = T^{*2}T^2 - 2T^*T + I = \begin{pmatrix} \varphi^2 - \varphi - 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus $T\beta_2(T) = 0$, so T is a 2-partial isometry.

Using $2 - \varphi^2 = 1 - \varphi = \delta$, we obtain

$$S = \begin{pmatrix} a\delta & 2 \\ 0 & 0 \end{pmatrix}.$$

Because $\delta\varphi = -1$,

$$P := ST = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q := TS = \begin{pmatrix} -1 & 2a \\ a\delta & 2 \end{pmatrix}. \quad (1)$$

In particular, $TST = T$ and $STS = S$, so S is indeed a generalized inverse.

Suppose, toward a contradiction, that S were the Moore–Penrose inverse for some Hilbert inner product. Relative to the original coordinates, this inner product is represented by a positive definite Hermitian matrix

$$G = \begin{pmatrix} x & z \\ \bar{z} & y \end{pmatrix}.$$

By Lemma 1, $P^*G = GP$. Using the first matrix in (1), this equality forces $z = 0$. Hence $G = \text{diag}(x, y)$ with $x, y > 0$.

The $(1, 2)$ entry of the remaining identity $Q^*G = GQ$ is

$$a\delta y = 2ax,$$

or, since $a > 0$,

$$\delta y = 2x.$$

This is impossible: $x, y > 0$ while $\delta < 0$. Therefore no positive definite G satisfies the two adjoint equations, and Lemma 1 proves the claim. $\square$

4 Scope, verification, and novelty

The theorem fully answers the universal yes/no Question 6 negatively. It does not classify which 2-partial isometries have a canonical generalized inverse that becomes Moore–Penrose after renorming, nor does it address the paper’s broader Questions 1–5.

The accompanying exact-arithmetic script verifies $T\beta_2(T) = 0$, the formula for S , both generalized-inverse identities, and the two linear equations in the entries of G . The script is a regression check and is not needed for the proof.

A bounded novelty search on 17 August 2026 covered the four run indexes, the current arXiv record (still version 1), exact searches for the title and Question 6, and variants combining 2-partial isometries, Moore–Penrose inverses, and changes of inner product. No later answer was found. Because the source appeared only in June 2026, novelty confidence is moderate pending specialist review.

Human review recommendation. Check the interpretation that the displayed $S = 2T^* - T^{*2}T$ is fixed while the inner product is varied, as the wording of Question 6 indicates. Then check the four products in (1) and the two G -selfadjointness equations; no external structural theorem is used.

References

- [1] M. A. Aouichaoui, M. Buchała, and S. R. Garcia, *On m -partial isometries: spectra, weighted shifts, and similarity*, arXiv:2606.12865v1 (2026).